ANALISI MATEMATICA 2 - LEZIONE 21

ALCUNI ESERCIZI DEL FOGLIO 5

1.a $f(x,y,z)=x+z$ Max/min in

$$D=\left\{(x,y,z): \begin{matrix} x^2+y^2+z^2=2 \\ z=x^2+y^2 \end{matrix}\right\}$$

$f\in C^2(\mathbb{R}^3)$, D è chiuso e limitato. Poniamo

$g_1(x,y,z)=x^2+y^2+z^2-2=0$, $g_2(x,y,z)=x^2+y^2-z=0$.

Regolarità:

$$\begin{bmatrix} g_{1x} & g_{1y} & g_{1z} \\ g_{2x} & g_{2y} & g_{2z} \end{bmatrix} = \begin{bmatrix} 2x & 2y & 2z \\ 2x & 2y & -1 \end{bmatrix}$$

Il rango non vale 2 se e solo se

$$\begin{cases} \det\begin{bmatrix} 2x & 2z \\ 2x & -1 \end{bmatrix} = 2x(-1-2z)=0 \\ \det\begin{bmatrix} 2y & 2z \\ 2y & -1 \end{bmatrix} = 2y(-1-2z)=0 \end{cases} \iff \begin{matrix} (0,0,z)\in D \\ (x,y,-\frac{1}{2})\in D \end{matrix}$$

Tutti i punti di D sono regolari.

Moltiplicatori:

$$\begin{cases} 1=\lambda 2x+\mu 2x \\ 0=\lambda 2y+\mu 2y \\ 1=\lambda 2z-\mu \\ x^2+y^2+z^2=2 \\ z=x^2+y^2 \end{cases}$$

$1=\lambda 2x+\mu 2x \to 1=2x(\lambda+\mu) \to \lambda+\mu\neq 0$

$0=\lambda 2y+\mu 2y \to 2y(\lambda+\mu)=0 \to y=0$ ($\lambda+\mu\neq 0$)

$$\begin{cases} y=0 \\ 1=2(\lambda+\mu)x \\ 1=2\lambda z-\mu \\ x^2+z^2=2 \\ z=x^2 \end{cases}$$

$x^2+z^2=2 \to z+z^2=2$

$z=1$ $\quad z=-2=x^2$

$x=1$ $\quad x=-1$ $\quad \emptyset$

Punti stazionari vincolati:

$(1,0,1)$ $\lambda=\frac{1}{2}$, $\mu=0$

$(-1,0,1)$ $\lambda=\frac{1}{6}$, $\mu=-\frac{2}{3}$

$f(1,0,1)=2$	Max
$f(-1,0,1)=0$	Min

$(1,0,1)$ punto di max assoluto

$(-1,0,1)$ punto di min assoluto

2.a $\displaystyle\iint_D \frac{e^{x/y}}{y^4}\,dx\,dy \qquad D=[0,2]\times[1,2]$

$$=\int_{y=1}^{2}\frac{1}{y^4}\left(\int_{x=0}^{2}e^{x/y}dx\right)dy=\int_{y=1}^{2}\frac{1}{y^4}\left[ye^{x/y}\right]_0^2dy=\int_1^2\frac{e^{2/y}-1}{y^3}dy$$

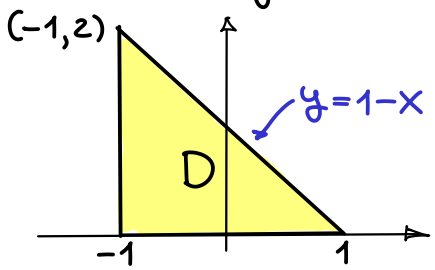
$t=\frac{1}{y}$, $dt=-\frac{dy}{y^2}$

$$=\int_1^{1/2}\frac{e^{2t}-1}{1/t}(-dt)=\int_{1/2}^{1}(te^{2t}-t)\,dt$$

$$=\left[\frac{te^{2t}}{2}-\frac{e^{2t}}{4}-\frac{t^2}{2}\right]_{1/2}^{1}=\frac{e^2}{2}-\frac{e^2}{4}-\frac{1}{2}-\left(\frac{e}{4}-\frac{e}{4}-\frac{1}{8}\right)=\frac{e^2}{4}-\frac{3}{8}$$

dove $\displaystyle\int te^{2t}dt=\int t\,d\left(\frac{e^{2t}}{2}\right)=\frac{te^{2t}}{2}-\frac{e^{2t}}{4}+c.$

2.b $\displaystyle\iint_D(1-|x|-y)\,dx\,dy \qquad D=\left\{(x,y):\begin{array}{l}x\geq -1,\ y\geq 0\\ x+y\leq 1\end{array}\right\}$



$$=\int_{x=-1}^{1}\left(\int_{y=0}^{1-x}(1-|x|-y)\,dy\right)dx$$

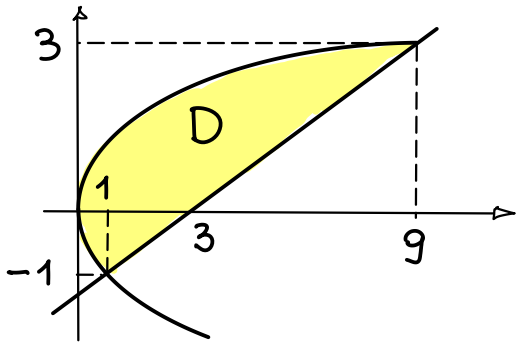
$$=\int_{-1}^{1}\left[(1-|x|)y-\frac{y^2}{2}\right]_0^{1-x}dx=\int_{-1}^{1}\left((1-|x|)(1-x)-\frac{1}{2}(-x)^2\right)dx$$

$1-|x|-x+x|x|-\frac{1}{2}+x-\frac{x^2}{2}$

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$$=2\int_0^1\left(\frac{1}{2}-x-\frac{x^2}{2}\right)dx=\left[x-x^2-\frac{x^3}{3}\right]_0^1=-\frac{1}{3}.$$

2.e $\iint_D \frac{1}{(2y-x+8)^2}\,dx\,dy \qquad D=\{(x,y): y^2\le x\le 2y+3\}$



$$=\int_{y=-1}^{3}\left(\int_{x=y^2}^{2y+3}\frac{1}{(2y-x+8)^2}\,dx\right)dy$$

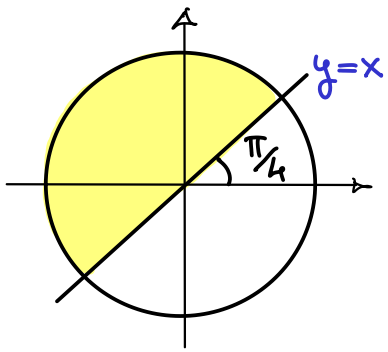
$$=\int_{y=-1}^{3}\left[\frac{1}{2y-x+8}\right]_{y^2}^{2y+3}dy=\int_{-1}^{3}\left(\frac{1}{5}-\frac{1}{2y-y^2+8}\right)dy$$

$-(y+2)(y-4)$

$$=\frac{3-(-1)}{5}+\frac{1}{6}\int_{-1}^{3}\left(\frac{1}{y-4}-\frac{1}{y+2}\right)dy=\frac{4}{5}+\frac{1}{6}\left[\log\left|\frac{y-4}{y+2}\right|\right]_{-1}^{3}$$

$$=\frac{4}{5}+\frac{1}{6}\left(\log\left(\frac{1}{5}\right)-\log(5)\right)=\frac{4}{5}-\frac{\log(5)}{3}.$$

2.f $\iint_D \sqrt{1-x^2-y^2}\,dx\,dy \qquad D=\{(x,y): y\ge x,\ x^2+y^2\le 1\}$



$$\overset{CP}{=}\int_{\rho=0}^{1}\sqrt{1-\rho^2}\left(\int_{\theta=\frac{\pi}{4}}^{\pi+\frac{\pi}{4}}d\theta\right)\rho\,d\rho$$

$t=1-\rho^2,\ dt=-2\rho\,d\rho$

$$=\pi\int_0^1\sqrt{1-\rho^2}\,\rho\,d\rho=\pi\int_0^1\sqrt{t}\cdot\frac{dt}{2}$$

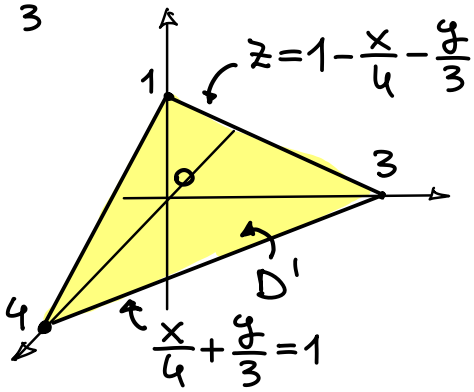
$$=\frac{\pi}{2}\left[\frac{2}{3}t^{3/2}\right]_0^1=\frac{\pi}{3}.$$

3.a $D=\left\{(x,y,z): \frac{|x|}{4}+\frac{|y|}{3}+|z|\le 1\right\}$ Volume?

D è simmetrico rispetto ai piani $x=0, y=0, z=0$ pertanto $|D|=8|E|$ dove

$$E=\left\{(x,y,z): \frac{x}{4}+\frac{y}{3}+z\le 1,\ x,y,z\ge 0\right\}$$

1) Per fili con $\varphi_1(x,y)=0$, $\varphi_2(x,y)=1-\frac{x}{4}-\frac{y}{3}$



e $D'=\left\{(x,y): \frac{x}{4}+\frac{y}{3}\le 1,\ x,y\ge 0\right\}$.

$$|D|=8|E|=8\iint_{D'}\left(1-\frac{x}{4}-\frac{y}{3}\right)dxdy$$

$$=8\int_{x=0}^{4}\left(\int_{y=0}^{3-\frac{3x}{4}}\left(1-\frac{x}{4}-\frac{y}{3}\right)dy\right)dx = 8\int_{u=0}^{1}\left(\int_{v=0}^{1-u}(1-u-v)3\,dv\right)4\,du$$

$\begin{cases}x=4u\\ y=3v\end{cases}$

$$=8\cdot 12\int_0^1\left[v-uv-\frac{v^2}{2}\right]_0^{1-u}du = 8\cdot 12\int_0^1\left((1-u)^2-\frac{(1-u)^2}{2}\right)du$$

$$=\frac{8\cdot 12}{2}\int_0^1(1-u)^2du = 48\left[-\frac{(1-u)^3}{3}\right]_0^1=\frac{48}{3}=16.$$

2) Per sezioni con $z\in[0,1]$ e $S_z=\left\{(x,y): \frac{x}{4}+\frac{y}{3}\le 1-z,\ x,y\ge 0\right\}$

triangolo rettangolo con cateti $4(1-z)$ e $3(1-z)$

$$|D|=8|E|=8\int_{z=0}^{1}\left(\iint_{S_z}1\,dxdy\right)dz$$

$$=8\int_0^1|S_z|\,dz = 8\int_0^1\frac{4\cdot 3(1-z)^2}{2}dz = 8\cdot 6\left[-\frac{(1-z)^3}{3}\right]_0^1$$

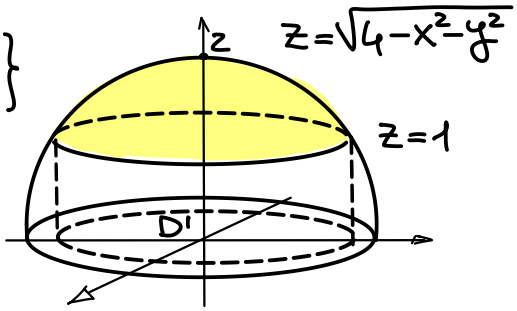
$$=8\cdot 2\cdot\frac{1}{3}=16.$$

3.b $D=\{(x,y,z): x^2+y^2+z^2\le 4, z\ge 1\}$ Volume?

↑ calotta

1) Per fili con $D'=\{(x,y): x^2+y^2+1\le 4\}$

$\varphi_1(x,y)=1$ e $\varphi_2(x,y)=\sqrt{4-x^2-y^2}$.



$$|D|=\iint_{D'}\left(\sqrt{4-x^2-y^2}-1\right)dxdy$$

$$\overset{CP}{=}\int_{\rho=0}^{\sqrt{3}}\int_{\theta=0}^{2\pi}\left(\sqrt{4-\rho^2}-1\right)\rho\,d\rho\,d\theta=2\pi\int_0^{\sqrt{3}}\left(\sqrt{4-\rho^2}-1\right)\rho\,d\rho$$

$t=4-\rho^2 \rightarrow dt=-2\rho d\rho$

$$=2\pi\int_4^1\left(t^{1/2}-1\right)\left(-\frac{dt}{2}\right)=\pi\left[\frac{2t^{3/2}}{3}-t\right]_1^4$$

$$=\pi\left(\left(\frac{16}{3}-4\right)-\left(\frac{2}{3}-1\right)\right)=\frac{5}{3}\pi.$$

2) Per sezioni con $z\in[1,2]$ e $S_z=\{(x,y): x^2+y^2\le 4-z^2\}$

↑ cerchio di raggio $\sqrt{4-z^2}$

$$|D|=\int_{z=1}^{2}\left(\iint_{S_z}1\,dxdy\right)dz=\int_1^2|S_z|\,dz$$

$$=\int_1^2\pi(4-z^2)\,dz=\pi\left[4z-\frac{z^3}{3}\right]_1^2$$

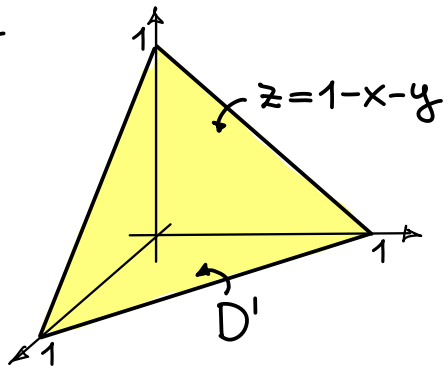
$$=\pi\left(8-\frac{8}{3}-4+\frac{1}{3}\right)=\frac{5}{3}\pi.$$

4.a $\displaystyle\iiint_D \frac{1}{(x+y+z+1)^3}\,dx\,dy\,dz$

dove $D=\{(x,y,z): x,y,z\geq 0,\ x+y+z\leq 1\}$.

Per fili con $\varphi_1(x,y)=0$, $\varphi_2(x,y)=1-x-y$

e $D'=\{(x,y): x,y\geq 0,\ x+y\leq 1\}$



$$=\iint_{D'}\left(\int_{z=0}^{1-x-y}\frac{1}{(x+y+z+1)^3}\,dz\right)dx\,dy$$

$$=\iint_{D'}\left[\frac{-1/2}{(x+y+z+1)^2}\right]_0^{1-x-y}dx\,dy=\frac{1}{2}\iint_{D'}\left(\frac{1}{(x+y+1)^2}-\frac{1}{4}\right)dx\,dy$$

$$=\frac{1}{2}\int_{x=0}^{1}\left(\int_{y=0}^{1-x}\frac{1}{(x+y+1)^2}\,dy\right)dx-\frac{|D'|}{8}=\frac{1}{2}$$

$$=\frac{1}{2}\int_0^1\left[-\frac{1}{x-y+1}\right]^{1-x}dx-\frac{1}{16}=\frac{1}{2}\int_0^1\left(\frac{1}{x+1}-\frac{1}{2}\right)dx-\frac{1}{16}$$

$$=\frac{1}{2}\left[\log(x+1)\right]_0^1-\frac{1}{4}-\frac{1}{16}=\frac{\log(2)}{2}-\frac{5}{16}.$$